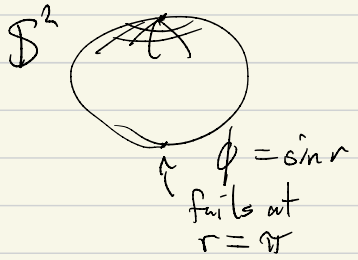
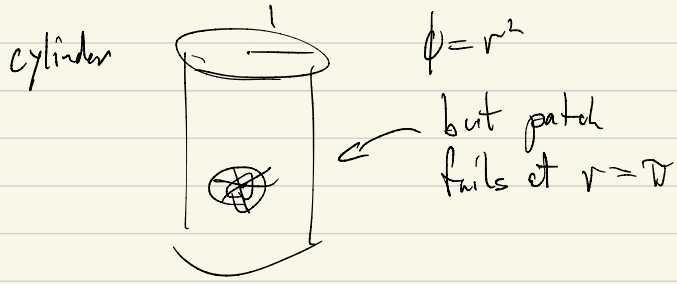
Minimizing length

Consider a geodesic polar patch $\sigma(r,\theta)$ centered at $p \in S$

$$ds^2 = dr^2 + \phi^2 d\theta^2 \qquad \phi = \phi(r,\theta) \sim r \text{ as } r \to 0$$

The patch is regular as long as σ is injective and $\phi > 0$





We assume geodesic polar patches are circular $\{0 < r < r_{max}\}$

Prop. Within a geodesic polar patch, the shortest path from p to any other pt. is the radial geodesic.

Proof: Suppose $q = \sigma(r_1, \theta_1)$ so the radial path p to q has length $r_1 < r_{max}$

Let η be a curve with $\eta(0) = p$, $\eta(1) = q$

If η stays in the patch then we can write $\eta(t) = (r(t), \theta(t))$

$$\begin{aligned} l(\eta) &= \int_0^1 \sqrt{\dot r^2 + \phi^2 \dot\theta^2}\, dt \\ &\geq \int_0^1 |\dot r|\, dt \\ &\geq \int_0^1 \dot r\, dt \\ &= r_1 \end{aligned}$$

If η leaves the patch, the same argument shows $l(\eta) \geq r_{max}$

Cor geodesics are locally length-minimizing.

Gauss's Theorema Egregium

Thm K depends only on g and so is invariant under local isometry

(Recall: $K = \frac{\det \mathrm{II}}{\det g}$ where $\mathrm{II}_{ij} = \sigma_{ij} \cdot N$, $g_{ij} = \sigma_i \cdot \sigma_j$)

Proof: For simplicity, assume $g_{12} = g_{21} = 0$, for example in a geodesic patch.

This means $\sigma_1 \perp \sigma_2$

Refine an orthonormal frame e_1, e_2, N

where $e_i := \frac{\sigma_i}{\sqrt{g_{ii}}}$

Note that all vectors depend on (u^1, u^2)

$e_i \cdot e_i = 1 \Rightarrow \partial_j e_i \perp e_i$ for all i, j

$\Rightarrow$ We can expand $\begin{cases} \partial_j e_1 = (\sim) e_2 + (\sim) N \\ \partial_j e_2 = (\sim) e_1 + (\sim) N \end{cases}$

Calculate the normal coeff,

$$\partial_j e_i = \frac{1}{\sqrt{g_{ii}}} \sigma_{ji} + (\sim) \sigma_i$$

$$\Rightarrow \partial_j e_i \cdot N = \frac{1}{\sqrt{g_{ii}}} \mathrm{II}_{ji}$$

$\therefore$

$$(\partial_1 e_1) \cdot (\partial_2 e_2) = \frac{1}{\sqrt{\det g}} \mathrm{II}_{11} \mathrm{II}_{22}$$

$$(\partial_1 e_2) \cdot (\partial_2 e_1) = \frac{1}{\sqrt{\det g}} \mathrm{II}_{12} \mathrm{II}_{21}$$

$$K = \frac{1}{\sqrt{\det g}} \left[(\partial_1 e_1) \cdot (\partial_2 e_2) - (\partial_1 e_2) \cdot (\partial_2 e_1) \right]$$

$$(\partial_1 e_1)\cdot(\partial_2 e_2) = \partial_2(\partial_1 e_1 \cdot e_2) - (\partial_{12} e_1)\cdot e_2$$

$$(\partial_1 e_2)\cdot(\partial_2 e_1) = \partial_1(e_2 \cdot \partial_2 e_1) - e_2\cdot(\partial_{12} e_1)$$

$$\therefore [\cdots] = \partial_2(\partial_1 e_1 \cdot e_2) - \partial_1(e_2 \cdot \partial_2 e_1)$$

$$\partial_1 e_1 \cdot e_2 = -\frac{1}{\sqrt{\det g}}\, \underbrace{\sigma_1 \cdot \sigma_{12}}_{= \frac{1}{2}\partial_2 g_{11}} \qquad (\text{since } \sigma_1 \perp \sigma_2)$$

$$e_2 \cdot \partial_2 e_1 = \frac{1}{\sqrt{\det g}}\, \underbrace{\sigma_2 \cdot \sigma_{21}}_{= \frac{1}{2}\partial_1 g_{22}}$$

This gives

$$K = -\frac{1}{2}\frac{1}{\sqrt{\det g}}\left[\partial_2\left(\frac{\partial_2 g_{11}}{\sqrt{\det g}}\right) + \partial_1\left(\frac{\partial_1 g_{22}}{\sqrt{\det g}}\right)\right]$$

for $g_{12} \equiv 0$

If we write $g_{ii} = \phi_i^2$ this becomes

$$K = -\frac{1}{\phi_1\phi_2}\left[\partial_2\left(\frac{\partial_2 \phi_1}{\phi_2}\right) + \partial_1\left(\frac{\partial_1 \phi_2}{\phi_1}\right)\right]$$

Examples

1) S^2 $\quad ds^2 = dr^2 + \sin^2 r\, d\theta^2 \qquad \phi_1 = 1,\ \phi_2 = \sin r$

$$K = -\frac{1}{\sin r}\left[\partial_\theta(\cdots) + \partial_r\left(\frac{\partial_r \sin r}{1}\right)\right]$$

$$= 1$$

2) hyperbolic plane $\mathbb{H} := \{(x,y) : y > 0\}$

$$ds^2 = \frac{dx^2 + dy^2}{y^2}$$

$$\Rightarrow K = -1$$

Geodesic circles: $\{\text{radius} = r\}$ in a geodesic polar patch

$$ds^2 = dr^2 + \phi^2 d\theta^2 \qquad \phi \sim r \text{ as } r \to 0$$

$$K = -\frac{\partial_r^2 \phi}{\phi}$$

$$\Rightarrow K(p) = \lim_{r \to 0} \left(-\frac{\partial_r^2 \phi}{\phi} \right)$$

$$\phi = r + a_2 r^2 + a_3 r^3 + \cdots \qquad a_j = a_j(\theta) \text{ in principle}$$

$$K(p) = \lim_{r \to 0} \left(-\frac{2a_2 + 6a_3 r + \cdots}{r + a_2 r^2 + \cdots} \right)$$

limit exists $\Rightarrow a_2 = 0$ and $K(p) = -6a_3$

$$\therefore \phi = r + \frac{1}{6} K(p) r^3 + O(r^4)$$

$$\ell(\text{circle of radius } r) = \int_0^{2\pi} \phi(r, \theta)\, d\theta$$

$$= 2\pi r \left(1 - \frac{K(p)}{6} r^2 + O(r^3) \right)$$

$$\text{Area(disk of radius } r) = \int_0^r \int_0^{2\pi} \phi \, d\theta$$

$$= \pi r \left(1 - \frac{K(p)}{12} r^2 + O(r^3) \right)$$